What else can we do with lcWGS data?

This is not an exhaustive list…

Downstream applications of lcWGS data

Genotype–phenotype association

Genotype–environment associations

Introgression, local ancestry or gene flow analyses

Selection scans

Relatedness analyses

Species / population assignment

…and more — see the list at the end

Automated snakemake pipelines

Bioinformatics, 2025, **41(3)**, btaf105
https://doi.org/10.1093/bioinformatics/btaf105
Advance Access Publication Date: 11 March 2025
Applications Note

Genome analysis

PopGLen—a Snakemake pipeline for performing population genomic analyses using genotype likelihood-based methods

Zachary J. Nolen[1,*]

[1]Department of Biology, Lund https://orcid.org/0000-0001-8146-8016

*Corresponding author. Department of Biology, Lund University, Kontaktvägen 10, Lund 22 362, Sweden. E-mail: zachary.nolen@biol.lu.se.

Associate Editor: Peter Robinson

Abstract

Bioinformatics Advances, 2024, **00**, vbae098
https://doi.org/10.1093/bioadv/vbae098
Advance Access Publication Date: 11 July 2024
Application Note

Genetics and population analysis

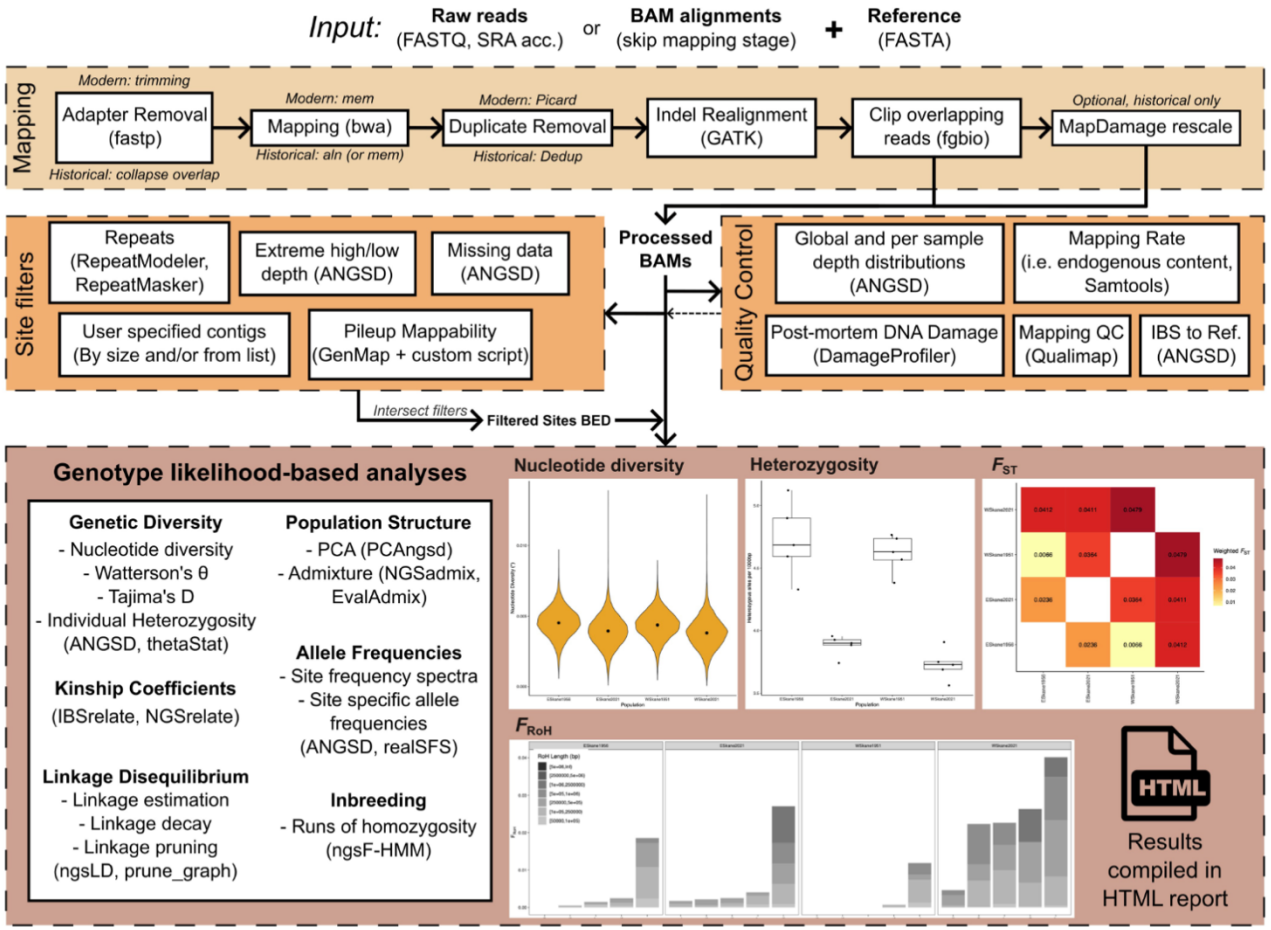
loco-pipe: an automated pipeline for population genomics with low-coverage whole-genome sequencing

Zehua T. Zhou[1], Gregory L. Owens[2], Wesley A. Larson[3], Runyang Nicolas Lou[1,*], Peter H. Sudmant[1,4,*]

[1]Department of Integrative Biology, University of California Berkeley, Berkeley, CA 94720, USA
[2]Department of Biology, University of Victoria, Victoria, BC V8P 5C2, Canada
[3]National Marine Fisheries Service, Alaska Fisheries Science Center, National Oceanographic and Atmospheric Administration, Auke Bay Laboratories, Juneau, AK 99801, USA
[4]Center for Computational Biology, University of California Berkeley, Berkeley, CA 94720, USA

*Corresponding author. Department of Integrative Biology, University of California, Berkeley, 4111 Valley Life Sciences Building, Harmon Way, Berkeley, CA 94720, USA. E-mail: nicolas931010@berkeley.edu (R.N.L.); Department of Integrative Biology, Center for Computational Biology, University of California, Berkeley, 4112 Valley Life Sciences Building, Harmon Way, Berkeley, CA 94720, USA. E-mail: psudmant@berkeley.edu (P.H.S.)

Associate Editor: Dr. Nicola Mulder



Types of input formats for downstream analyses

Many programs use allele counts or allele frequency tables as input

A. Minor allele frequency tables

Generated in R by joining the xxx.maf.gz files for each population into one table, with one MAF column per population.

Chr	Pos	MAF Pop1	MAF Pop2
1	10	0.4	0.1
1	20	0.3	0.2
2	50	0.1	0.7
2	80	0.8	0.4

Allele counts per site and population

B. Allele counts can be generated in multiple ways. In R, multiply the allele frequency (knownEM) column of the xxx.maf.gz file by the number of alleles at that site, 2 × nInd (number of genotyped individuals).

```
# Estimate minor allele count (x) and major allele count (y) for each population
JIGA.maf$x <- round(JIGA.maf$knownEM * (2 * JIGA.maf$nInd), digits = 0)
JIGA.maf$y <- (2 * JIGA.maf$nInd) - JIGA.maf$x
```

The per-population counts can then be joined into a single table with one column per population.

Downside: genotype uncertainty is not carried over. But with sufficient sample sizes, allele frequencies inferred from lcWGS data are generally very accurate.

Part 2

Approaches that can use these data

Population-level GWAS and GEA with BayPass

Input: allele counts — one row per SNP, two columns (allele 1, allele 2) per population

```
      Pop1    Pop2
--- file begins here ---
snp1  81 19   86 14   2 98 8 92 32 68 23 77
snp2  89 11   81 19   9 91 1 99 27 73 27 73
snp3  89 11   91 9    0 0 15 85 77 23 80 20

[...97 more lines...]

--- file ends here ---
```

Pop1 and **Pop2** columns highlighted; each population contributes a pair of allele counts per SNP.

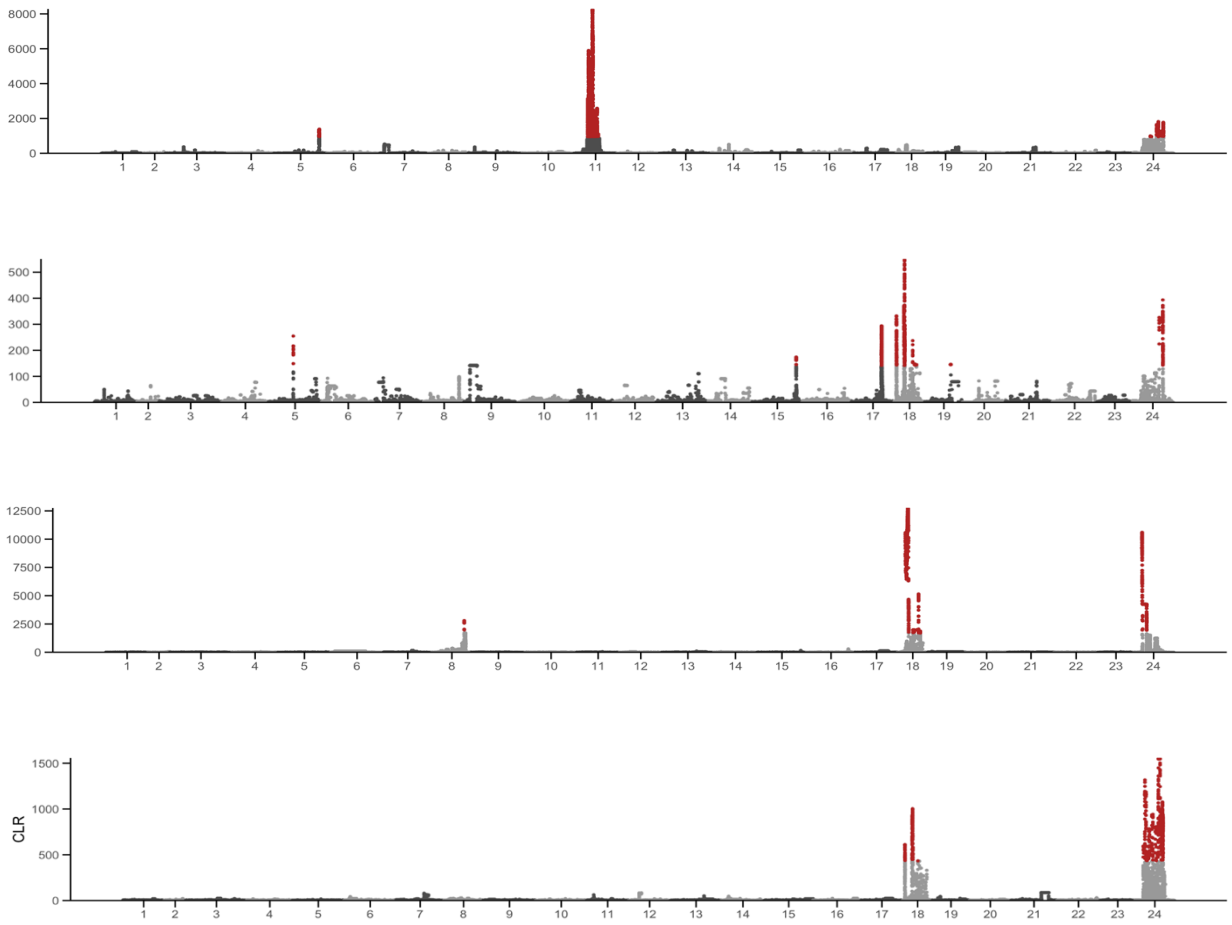
Selective sweeps from minor or derived allele counts

position	x	n	folded
460000	9	100	0
460010	100	100	0
460210	30	78	1
463000	0	94	0
...	...	...	...

x = allele count
n = number of samples
folded = polarised or not? (1 = not polarised)

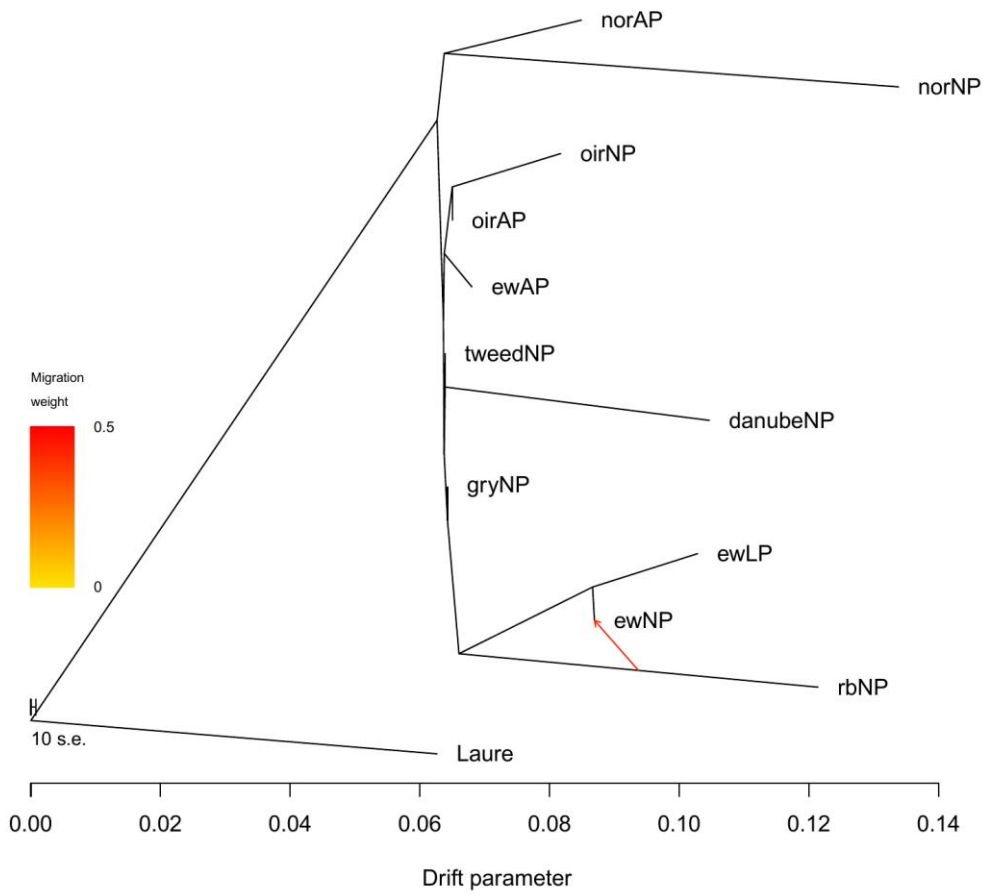
1. Estimate the SFS for the entire genome

2. Run SweepFinder2 or SweeD by chromosome for the entire genome



CLR along chromosomes 1–24

Phylogenetic relationships and gene flow: TreeMix



TreeMix uses a simple allele-count table as input (counts per population, comma-separated):

```
pop1 pop2 pop3 pop4
5,1  1,1  4,0  0,4
3,3  0,2  2,2  0,4
1,5  0,2  2,2  1,3
```

Pickrell & Pritchard 2012, PLoS Genetics

Recombination rate inference: ReLERNN

Predicting the Landscape of Recombination Using Deep Learning

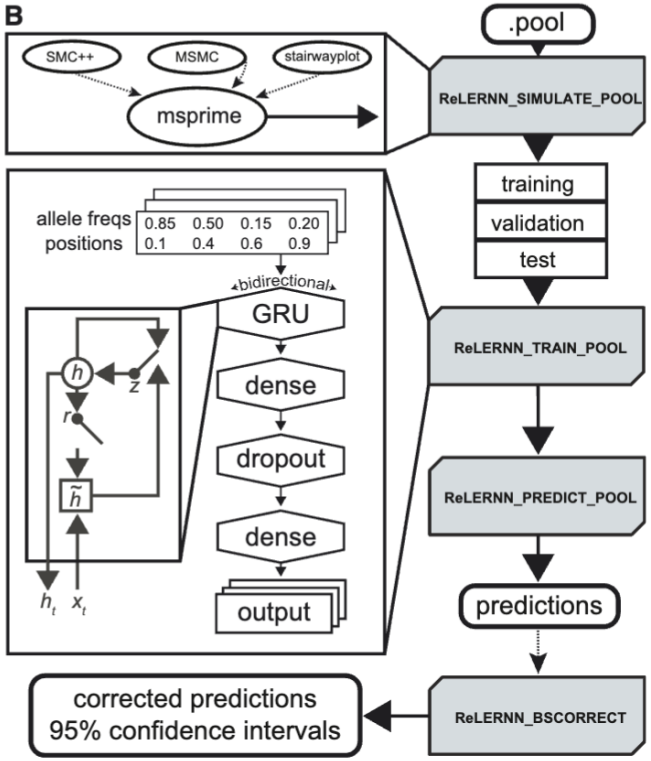
Jeffrey R. Adrion,*†,[1] Jared G. Galloway,†,[1] and Andrew D. Kern [1]
[1]Institute of Ecology and Evolution, University of Oregon, Eugene, OR
†These authors contributed equally to this work.
ReLERNN is currently available at https://github.com/kern-lab/ReLERNN.
***Corresponding author:** E-mail: jadrion@uoregon.edu.
Associate editor: Claus Wilke

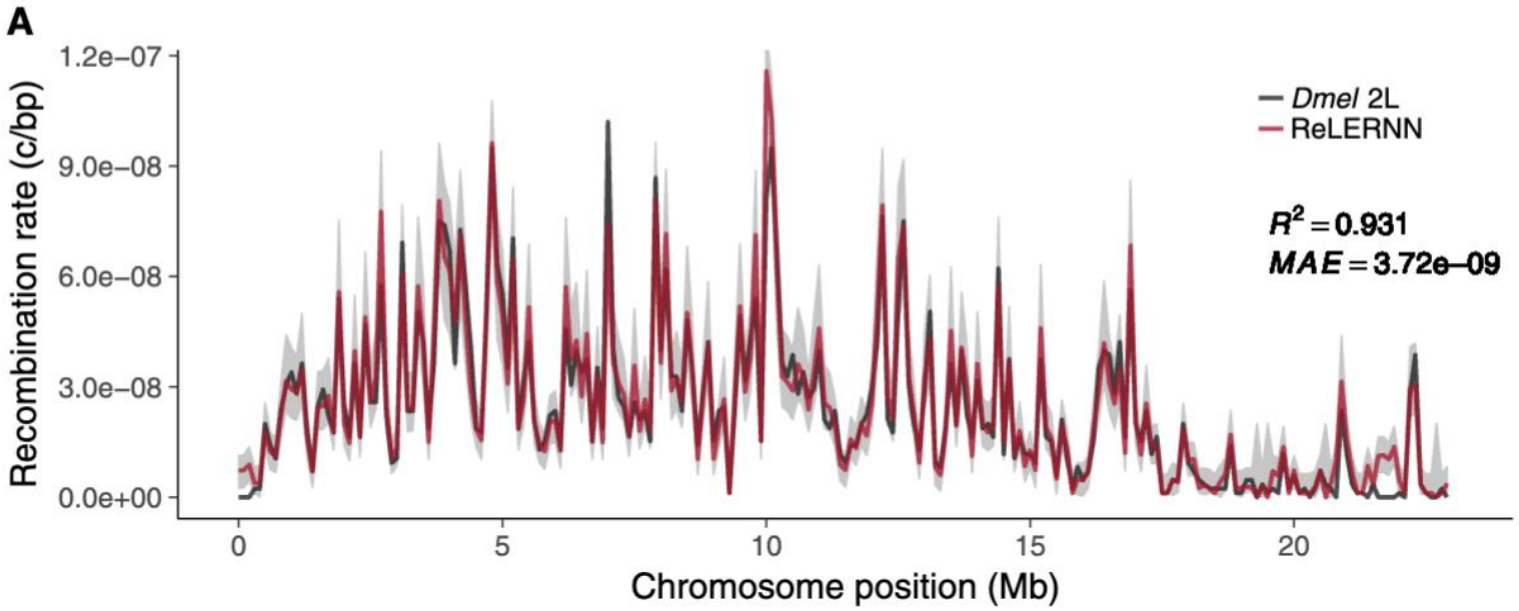
OXFORD GENETICS
GENETICS, 2025, iyaf108
https://doi.org/10.1093/genetics/iyaf108
Advance Access Publication Date: 5 June 2025
Investigation

Estimating recombination using only the allele frequency spectrum

Matthew W. Hahn [1,2,*] Sarthak R. Mishra [2]
[1]Department of Biology, Indiana University Bloomington, 1001 E. 3rd St., Bloomington, IN 47405, USA
[2]Department of Computer Science, Indiana University Bloomington, 700 N. Woodlawn Ave., Bloomington, IN 47405, USA
*Corresponding author: Department of Biology, Indiana University Bloomington, 1001 E. 3rd St., Bloomington, IN 47405, USA. Email: mwh@iu.edu

ReLERNN can use minor allele frequencies as input to infer recombination rate landscapes across the genome using recurrent neural networks.





Adrion et al.2020, MBE

Part 3

Genome-wide association studies

GWAS in ANGSD: -doAsso

```
./angsd -doAsso
abcAsso.cpp:
        -doAsso 0
        1: Frequency Test (Known Major and Minor)
        2: Score Test
        4: Latent genotype model
        5: Score Test with latent genotype model - hybrid test
        6: Dosage regression
        7: Latent genotype model (wald test) - NOT PROPERLY TESTED YET!
  Frequency Test Options:
        -yBin           (null)  (File containing disease status)

  Score, Latent, Hybrid and Dosage Test Options:
        -yBin           (null)  (File containing disease status)
        -yCount         (null)  (File containing count phenotypes)
        -yQuant         (null)  (File containing phenotypes)
        -cov            (null)  (File containing additional covariates)
        -sampleFile             (null)  (.sample File containing phenotypes and covariates)
        -whichPhe       (null)  Select which phenotypes to analyse, write phenos comma seperated ('phe1,phe2,...'), only works with a .sample
file
        -whichCov       (null)  Select which covariates to include, write covs comma seperated ('cov1,cov2,...'), only works with a .sample
file
        -model  1
        1: Additive/Log-Additive (Default)
        2: Dominant
        3: Recessive

        -minHigh        10      (Require atleast minHigh number of high credible genotypes)
        -minCount       10      (Require this number of minor alleles, estimated from MAF)
        -assoThres      0.000001        Threshold for logistic regression
        -assoIter       100     Number of iterations for logistic regression
        -emThres        0.000100        Threshold for convergence of EM algorithm in doAsso 4 and 5
        -emIter 40      Number of max iterations for EM algorithm in doAsso 4 and 5

        -doPriming      1       Prime EM algorithm with dosage derived coefficients (0: no, 1: yes - default)

        -Pvalue 0       Prints a P-value instead of a likelihood ratio (0: no - default, 1: yes)

  Hybrid Test Options:
        -hybridThres            0.050000        (p-value value threshold for when to perform latent genotype model)
```

Association tests

1 Frequency test
2 Score test
4 Latent genotype model
5 Hybrid (score + latent)
6 Dosage regression

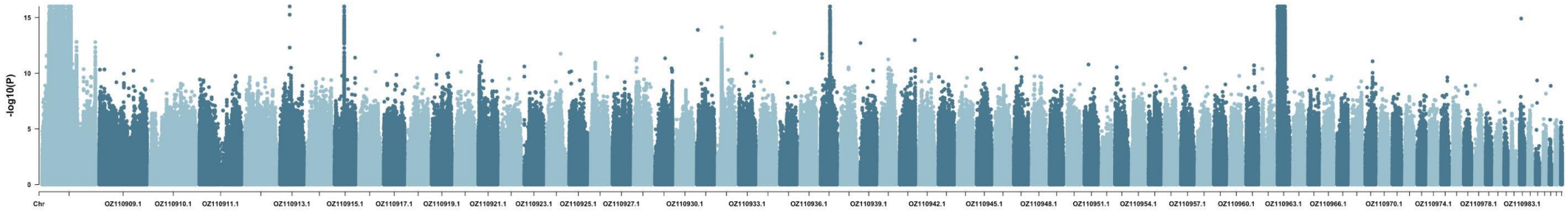
-model

1 additive · 2 dominant · 3 recessive

Covariates (e.g. PCs) via -cov

Genotype–phenotype association in lamprey

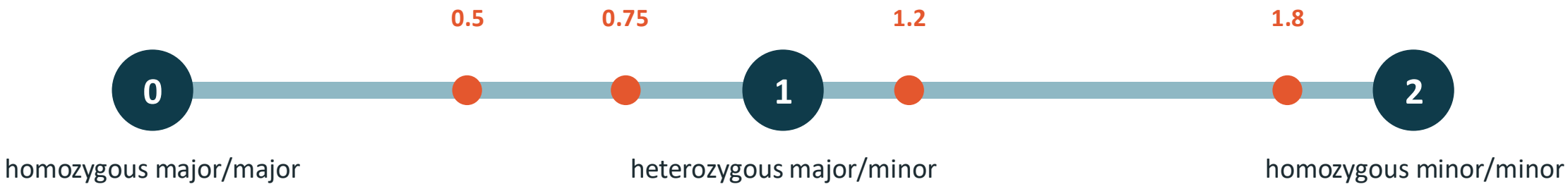
GWAS for life history using ANGSD, correcting for population structure with PC1–3.
Model: score test with latent genotype model (hybrid test, -doAsso 5).



Mean genotypes (allele dosages)

More sophisticated GWAS software can use mean genotypes as input, which retain some of the genotype uncertainty.

A mean genotype is a composite score of the genotype likelihoods (from the beagle file), ranging from 0 to 2.



Mean genotypes can take any intermediate value (e.g. 0.5, 0.75, 1.2, 1.8) — this captures the uncertainty of the genotype.

Estimating the mean genotype

Treat GLs as genotype probabilities; the expected genotype (dosage) is
0·P(AA) + 1·P(AB) + 2·P(BB) = P(AB) + 2·P(BB)

Marker	Allele 1	Allele 2	Ind0 (AA)	Ind0 (AB)	Ind0 (BB)	Ind1 (AA)	Ind1 (AB)	Ind1 (BB)
1_14000023	1	0	0.941177	**0.058822**	**0.000001**	0.799685	0.199918	0.000397
1_14000113	0	2	0.855993	0.106996	0.037010	0.333333	0.333333	0.333333
1_14000202	2	0	0.835380	0.104420	0.060201	0.799685	0.199918	0.000397

1. If all three GLs are 0.3333 (no information), set the site to NA

2. Allele dosage = P(Major/Minor) + 2 × P(Minor/Minor)

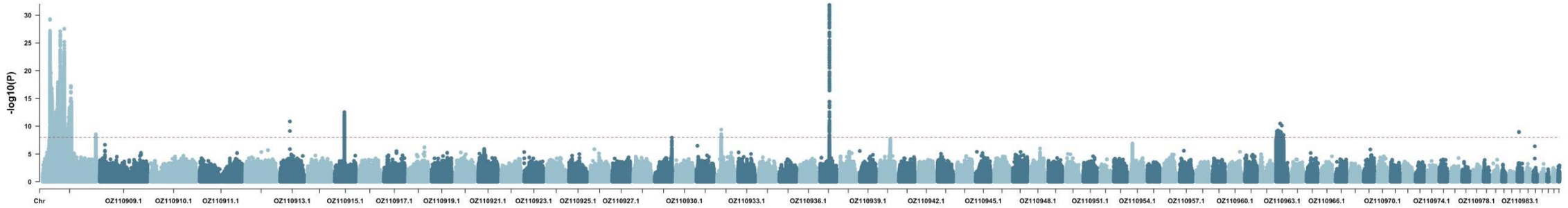
Example (Ind0, first site)
0.058822 + 2 × **0.000001** = 0.058824

Note: we also polarise by a random reference individual so the same minor allele is always used.

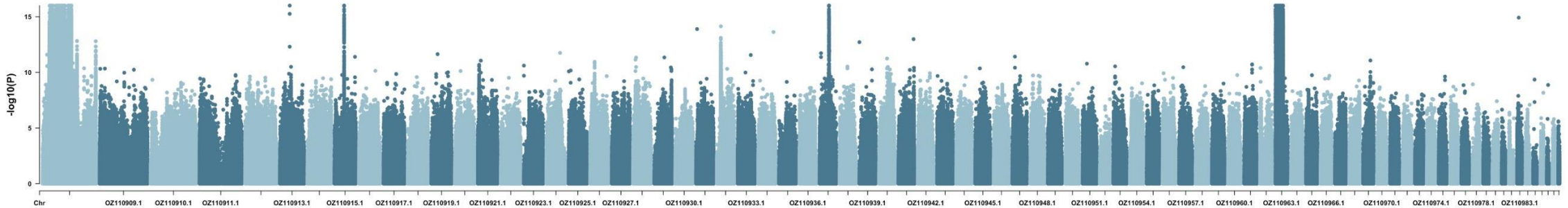
Mean genotypes in GEMMA vs ANGSD

Mean-genotype files can be used for linear mixed models in GEMMA, correcting with a genetic covariance (relatedness) matrix and principal components.

GWAS with GEMMA



GWAS with ANGSD



GEMMA: Zhou & Stephens 2012, Nature Genetics

Part 4

Demographic inference, gene flow and ancestry

Demographic inference

Option 1 · SFS-based

Use the 1D, 2D (or 3D…) SFS from ANGSD or winSFS with demographic inference software, e.g. ∂a∂i, GADMA, fastsimcoal2, Stairway Plot 2.

Important: the SFS can be problematic with very low coverage data (< 2×), which can lead to misinference. Newer approaches such as winSFS may help.

Do not apply a minor allele frequency filter! You want to keep all sites in the genome (but might want to remove coding sites)

Option 2 · ABC

Calculate summary statistics from lcWGS data and use them for demographic inference with approximate Bayesian computation.

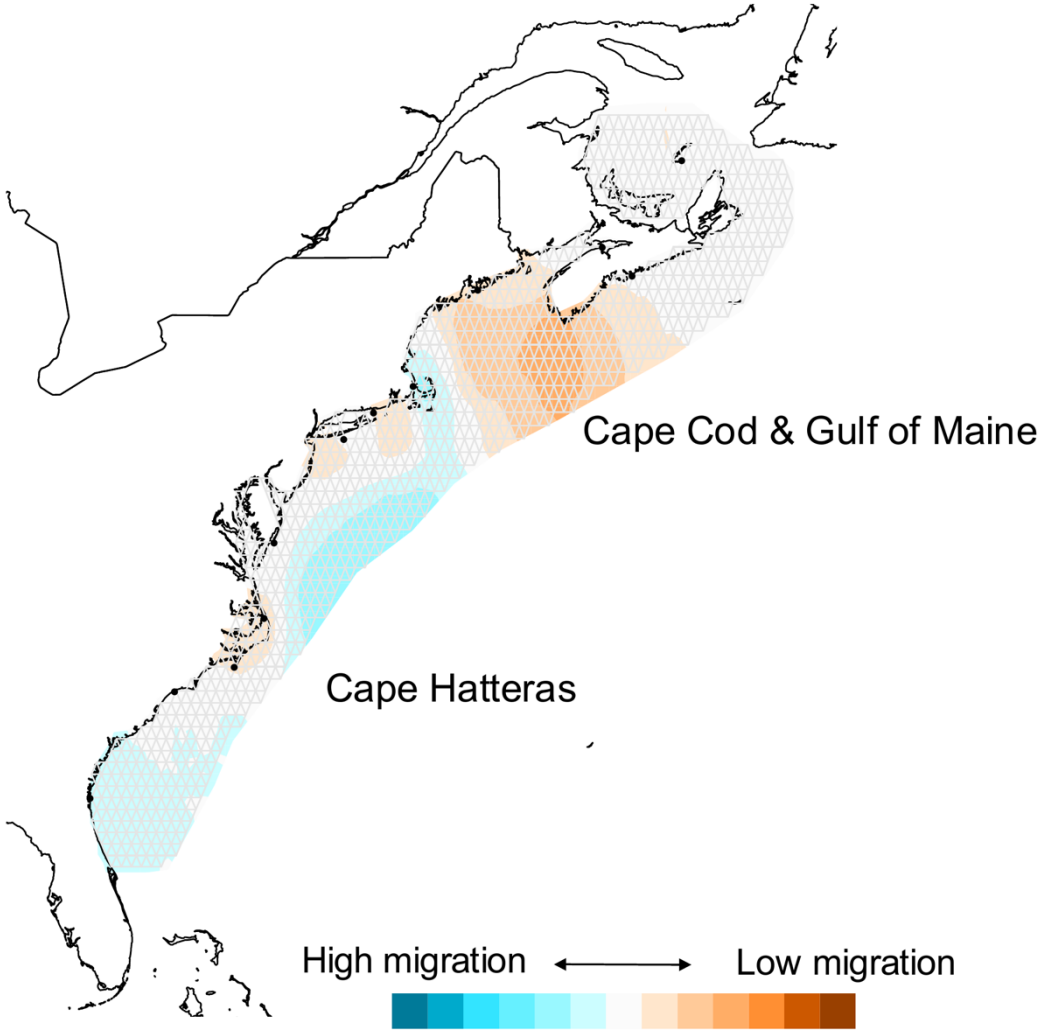
Disclaimer: not tested here; would require custom implementation.

Effective migration surfaces (EEMS)

Input

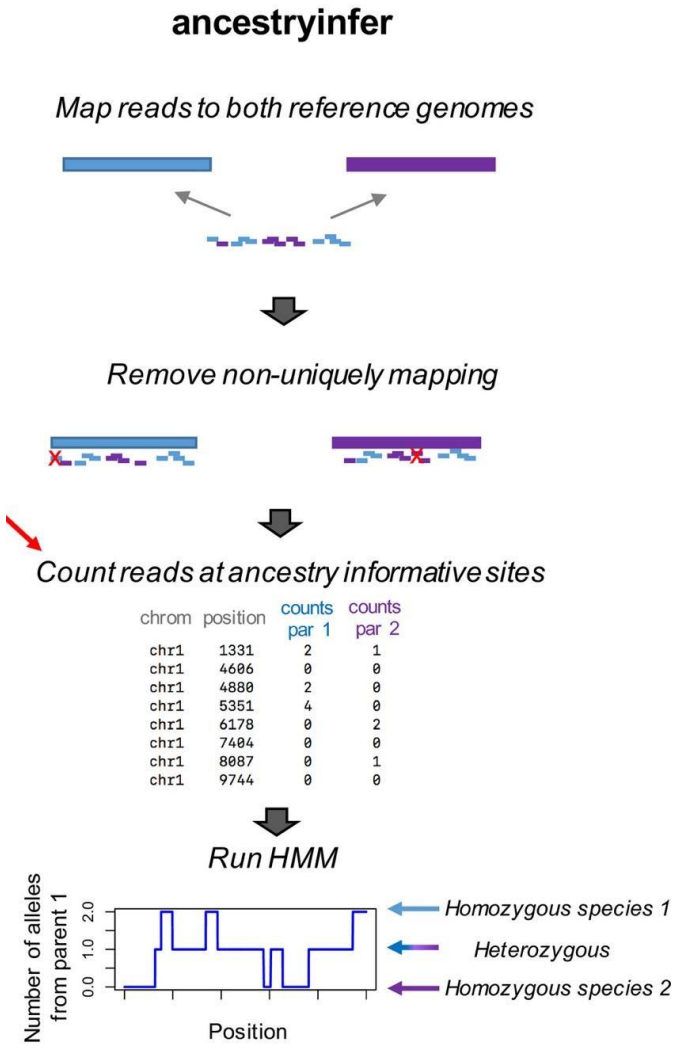
- Pairwise FST matrix
- IBS matrix from ANGSD

Posterior mean migration rates (log10 scale) along the US Atlantic coast: reduced effective migration around Cape Cod & the Gulf of Maine and near Cape Hatteras.



EEMS: Petkova et al. 2016, Nature Genetics

Local ancestry: ancestryinfer



1. Map reads to both reference genomes (or single genome)
2. Remove non-uniquely mapping reads
3. Count reads at ancestry-informative sites
4. Run the HMM to infer local ancestry

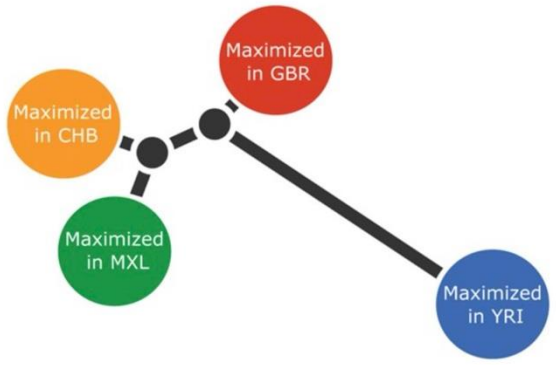
Input
Reference genomes for each species/population, plus fastq files for individuals.

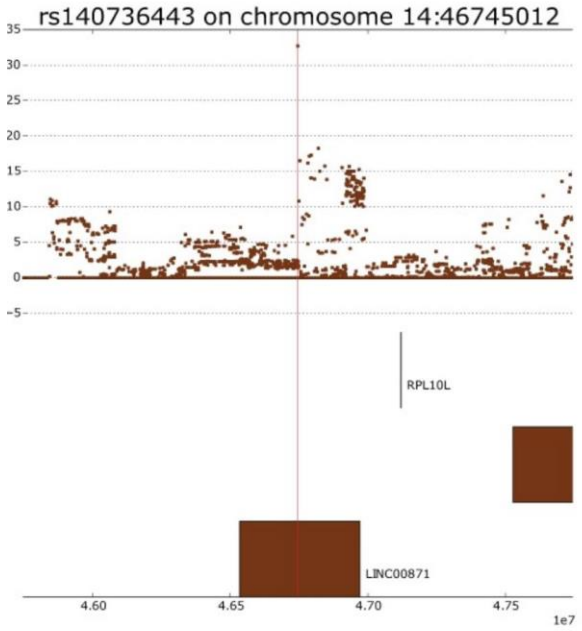
Caveat
Requires ancestry-informative SNPs which can be inferred using Fst or allele frequency differentiation.

Schumer et al. 2020, Molecular Ecology Resources

Selective sweeps across ancestry components: Ohana

Fig. 4.





JOURNAL ARTICLE

Detecting Selection in Multiple Populations by Modeling Ancestral Admixture Components

Jade Yu Cheng ✉, Aaron J Stern, Fernando Racimo, Rasmus Nielsen

Molecular Biology and Evolution, Volume 39, Issue 1, January 2022, msab294,
https://doi.org/10.1093/molbev/msab294
Published: 09 October 2021

Input: Beagle genotype likelihood file

1. Reconstruct population structure
2. Correct allele frequencies using the inferred structure
3. Scan for signatures of selection in specific ancestry clusters

Cheng et al. 2022, Molecular Biology and Evolution

Population assignment: WGSassign

< 0.5×

Super low-coverage WGS data is enough to assign individuals to source species or populations, using genotype likelihoods directly.

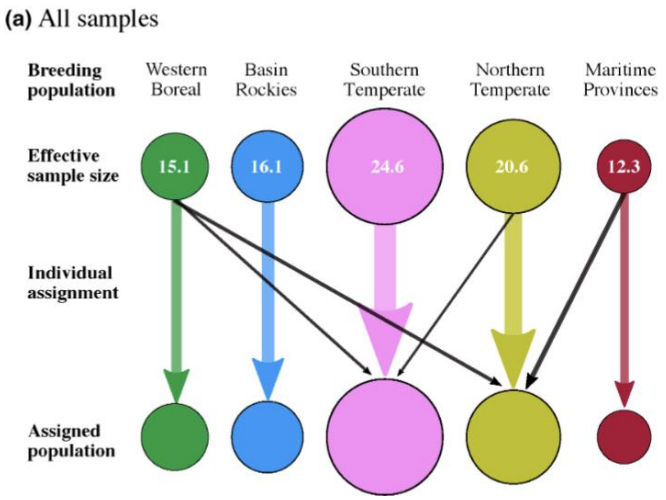
Received: 19 September 2023 | Accepted: 15 December 2023
DOI: 10.1111/2041-210X.14286

Methods in Ecology and Evolution

RESEARCH ARTICLE

Population assignment from genotype likelihoods for low-coverage whole-genome sequencing data

Matthew G. DeSaix[1] | Marina D. Rodriguez[1] | Kristen C. Ruegg[1] | Eric C. Anderson[1,2,3]

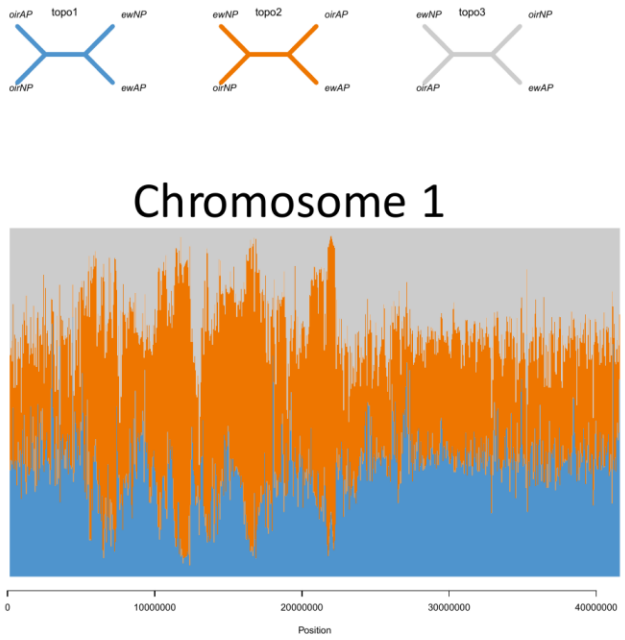


DeSaix et al. 2024, Methods in Ecology and Evolution

Phylogenetic relationships across the genome

Topology weighting with TWISST on IBS-based NJ trees

Which tree topology is most common in specific genomic regions?



1. Estimate an IBS matrix in windows across the genome (iterate over windows); window size can be set from LD decay

2. Build a neighbour-joining tree from each IBS matrix with the BIONJ algorithm, e.g. ape in R

3. Run TWISST to compute topology weightings from all trees across the genome

This relies on some custom scripts.

Jacobs et al. 2026 bioRxiv
Wang et al. 2024 Systematic Biology

Martin & Van Belleghem 2017, Genetics

Relatedness

Bioinformatics, 31(24), 2015, 4009–4011
doi: 10.1093/bioinformatics/btv509
Advance Access Publication Date: 30 August 2015
Applications Note

Sequence analysis

NgsRelate: a software tool for estimating pairwise relatedness from next-generation sequencing data

Thorfinn Sand Korneliussen[1] and Ida Moltke[2,*]

[1]Center for GeoGenetics and [2]Department of Biology, University of Copenhagen, 2200 Copenhagen, Denmark

*To whom correspondence should be addressed.
Associate Editor: John Hancock

Received on June 23, 2015; revised on August 6, 2015; accepted on August 24, 2015

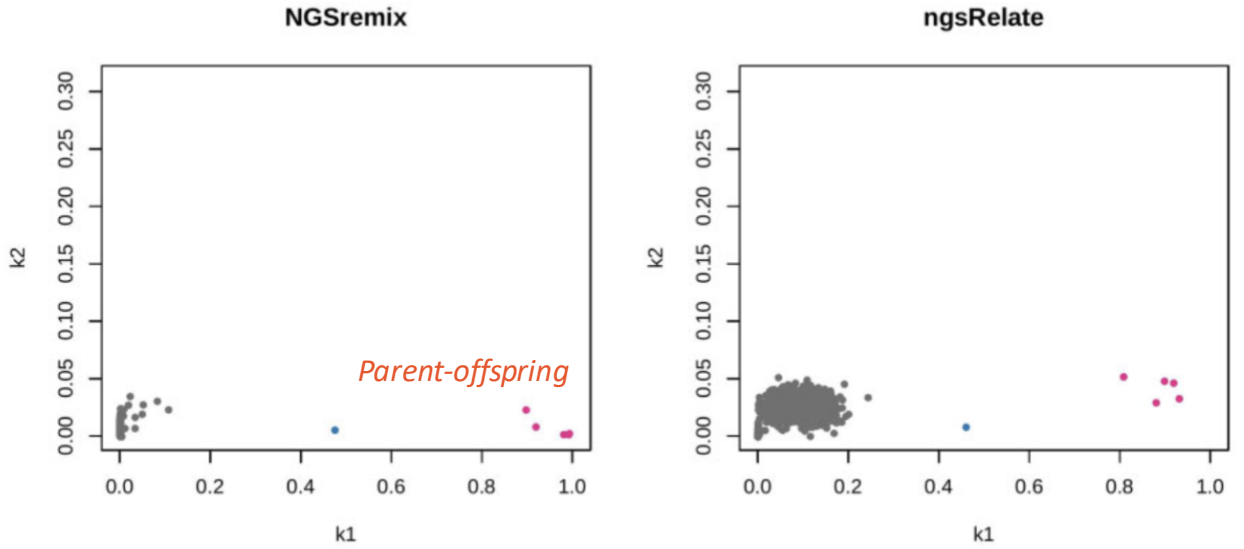
G3, 2021, **11(8)**, jkab174
DOI: 10.1093/g3journal/jkab174
Advance Access Publication Date: 20 May 2021
Software and Data Resources

NGSremix: a software tool for estimating pairwise relatedness between admixed individuals from next-generation sequencing data

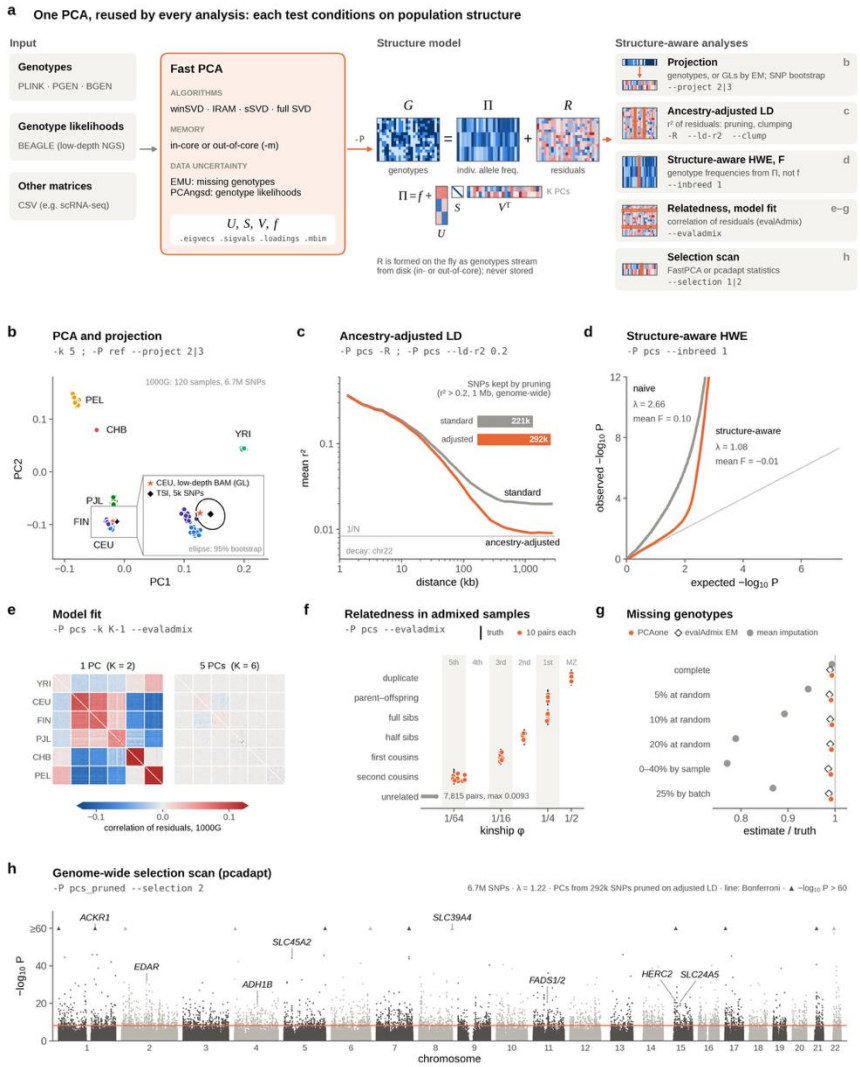
Anne Krogh Nøhr, [1,2,†] Kristian Hanghøj, [1,†] Genís Garcia-Erill, [1] Zilong Li, [1] Ida Moltke, [1,*] and Anders Albrechtsen[1,*]

[1]Department of Biology, The Bioinformatics Centre, University of Copenhagen, 2200 Copenhagen N, Denmark and
[2]H. Lundbeck A/S, 2500 Valby, Denmark
[†] These authors contributed equally to this work.

1. ngsRelate uses GLs in beagle format and allele frequencies as input



PCA, LD and relatedness in structured populations



1 PCAone is a new package that can use GLs to perform diverse analyses in structured populations

OXFORD GENETICS

GENETICS, 2025, 229(3), iyaf009
https://doi.org/10.1093/genetics/iyaf009
Advance Access Publication Date: 5 February 2025
Investigation

Measuring linkage disequilibrium and improvement of pruning and clumping in structured populations

Ulises Bercovich [1,†] Malthe Sebro Rasmussen [2,†] Zilong Li [2] Carsten Wiuf [1] Anders Albrechtsen [2,*]

[1]Department of Mathematical Sciences, University of Copenhagen, Copenhagen 2100, Denmark
[2]Department of Biology, University of Copenhagen, Ole Maaløes Vej 5, Copenhagen 2200, Denmark

*Corresponding author: Department of Biology, University of Copenhagen, Ole Maaløes Vej 5, Copenhagen 2200, Denmark. Email: aalbrechtsen@bio.ku.dk
†These authors contributed equally to this work.

Caveat
I have not tested this yet and cannot comment on how well it works.

Other approaches possible with lcWGS data

Ancestry / gene flow

- D-stats / ABBA-BABA (angsd - doAbbababa)
- TreeMix
- EEMS
- TWISST
- WGSassign

Selection

- PCAngsd (e.g. pcadapt)
- SweepFinder2
- Ohana
- BayPass
- Parallel AF change (AFVaper)
- AF differentiation

Relatedness and Pedigree

- ngsRelate
- LowKi
- NGSremix
- APOH (recent hybrid pedigree)

Genomic / geographic clines

- BGC-HM (based on GLs)
- HZAR (e.g. mean genotypes)

Linkage mapping

- Lep-MAP3

Most of these take allele counts/frequencies, mean genotypes, IBS/FST matrices or beagle GLs — the formats covered in this session.